Seminars 11 & 12: Computing the Spectrum

Topics: CTFT–DTFT–DFT relations, sampling, frequency indexing, DFT/IDFT, FFT complexity, windowing, spectral leakage, resolution, bin alignment, zero-padding, periodic signals, FIR spectra, and spectrograms.

> **Key idea.** An FFT computes the DFT exactly for the finite sequence supplied to it. Correct spectrum interpretation therefore requires separating the effects of sampling, finite-duration windowing, and evaluation on a discrete frequency grid.

1. From the Fourier integral to the DFT

A continuous-time signal is sampled with sampling period T_s, producing $x[n] = x_c(nT_s), n = 0, \dots, N-1$. The continuous-time Fourier transform is $X_c(j\omega) = \int_{-\infty}^{\infty} x_c(t)e^{-j\omega t}\, dt$.

(a) Approximate the Fourier integral using a finite Riemann sum based on the N available samples.

(b) Choose the analysis frequencies $\omega_k = \frac{2\pi k}{NT_s}$, $\quad k = 0, \dots, N-1$,

(c) and prove that the resulting expression is proportional to the DFT $X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$.

(d) Explain the role of T_s in converting the DFT coefficients into an approximation of the CTFT.

(e) Identify three distinct sources of discrepancy between an FFT-computed spectrum and the ideal CTFT.

2. The four Fourier representations

Complete a comparison of the continuous-time Fourier transform, discrete-time Fourier transform, Fourier series, and discrete Fourier transform.

(a) For each representation, state whether time is continuous or discrete.

(b) State whether frequency is continuous or discrete.

(c) State whether the frequency representation is periodic.

(d) State whether the signal is assumed finite, periodic, or potentially infinite.

(e) Write the corresponding analysis equation.

(f) Explain why all four representations can be interpreted as projections of a signal onto complex exponentials.

3. Sampling, DTFT periodicity, and aliasing

The continuous-time signal $x_c(t) = 2\cos(2\pi\, 600t) + \cos(2\pi\, 1300t)$
is sampled at $f_s = 2000$ samples/s.

(a) Determine the normalized radian frequencies of both sinusoidal components.

(b) Determine whether either component violates the sampling theorem.

(c) Find the aliases inside $0 \le f \le f_s/2$.

(d) Sketch one period of the magnitude DTFT of the sampled signal.

(e) Predict the DFT-bin locations of the peaks for $N = 100$.

(f) Repeat the analysis for $f_s = 4000$ samples/s and explain the difference.

4. Direct calculation of a short DFT and IDFT

Consider the four-sample sequence $x[n] = \{1,1,0,0\}$, $\quad 0 \le n \le 3$.

(a) Calculate its four-point DFT directly from the definition.

(b) Express every coefficient in rectangular and polar form.

(c) Verify conjugate symmetry.

(d) Use the inverse DFT to reconstruct $x[n]$.

(e) Identify the physical frequency represented by each bin when $f_s = 8000$ samples/s.

(f) Explain why bins $k = 3$ and $k = -1$ represent the same discrete-time frequency.

5. DFT bins and negative frequencies

A real signal is sampled at $16\,000$ samples/s and analysed with a 16-point DFT.

(a) Calculate the frequency spacing Δf.

(b) List the physical frequency represented by every bin $k = 0, \dots, 15$, using negative-frequency labels for the upper half.

(c) Identify the bins representing DC, the Nyquist frequency, $+3000$ Hz, and -3000 Hz.

(d) If $X[3] = 4 - 2j$, determine $X[13]$.

(e) State which DFT coefficients are necessarily real for an even-length DFT of a real signal.

(f) Explain why plotting all bins from 0 to f_s without relabelling the upper half can be misleading.

6. Bin-centred and non-bin-centred sinusoids

Let $x[n] = \cos\left(\frac{2\pi f_0 n}{f_s}\right)$, with $f_s = 1024$ samples/s and $N = 256$. Analyse the cases $f_0 = 128$ Hz and $f_0 = 130$ Hz.

(a) Calculate the DFT-bin spacing.

(b) Determine whether each sinusoid is bin-centred.

(c) Predict the positive- and negative-frequency peak locations.

(d) Explain why one case ideally produces energy in only two bins while the other spreads energy over many coefficients.

(e) Describe the relationship between the non-bin-centred DFT samples and a shifted Dirichlet-function envelope.

(f) Explain why this spreading is not an FFT error.

7. Rectangular-window spectrum and spectral leakage

A sinusoid is observed only over the interval $0 \le t < T$, which is equivalent to multiplying it by a rectangular window.

(a) Derive the CTFT of the duration-T rectangular window.

(b) Determine the locations of the first zeros.

(c) Show that the mainlobe width between the first zeros is $\Delta\omega = \frac{4\pi}{T}$.

(d) Describe how the ideal spectrum of an infinite-duration sinusoid changes after windowing.

(e) Explain the origin of spectral leakage.

(f) Determine how the mainlobe width changes if the observation duration is doubled.

(g) Explain why sidelobes may be mistaken for weak sinusoidal components.

8. Rectangular versus Hamming windows

Two equal-amplitude sinusoids have frequencies 1000 Hz and 1080 Hz and are sampled at $f_s = 8000$ samples/s. Use the approximate first-zero mainlobe widths $\Delta f_{\mathrm{rect}} \approx \frac{2}{T}$, $\Delta f_{\mathrm{Hamming}} \approx \frac{4}{T}$.

(a) Estimate the minimum observation duration required to resolve the two sinusoids with a rectangular window.

(b) Repeat for a Hamming window.

(c) Explain why the Hamming window needs a longer duration for comparable frequency resolution.

(d) State which window is preferable when detecting a weak sinusoid close to a strong sinusoid.

(e) State which is preferable when separating two nearby components of comparable amplitude.

(f) Discuss the trade-off between mainlobe width and sidelobe level.

9. Resolving a windowed sum of sinusoids

Consider $x_c(t) = 1 + 0.8\cos(2\pi\, 200t) + 0.15\cos(2\pi\, 230t)$, sampled at $f_s = 2000$ samples/s.

(a) Draw the ideal two-sided CTFT line spectrum.

(b) Predict the spectrum after observation with a 20-ms rectangular window.

(c) Repeat for a 100-ms rectangular window.

(d) Determine in which case the 200-Hz and 230-Hz components are likely to be resolved.

(e) Discuss whether the weak 230-Hz component may be obscured by leakage from the 200-Hz component.

(f) Explain how a Hamming window changes the result.

(g) Distinguish failure to resolve components from insufficient DFT-bin density.

10. Zero-padding and true frequency resolution

A length-$L = 40$ sinusoidal sequence is analysed using: (a) a 40-point DFT; (b) a 256-point DFT after appending zeros; and (c) a 256-point DFT of 256 actual signal samples.

(a) Compare the DFT-bin spacings.

(b) Identify which two analyses have the same window mainlobe width.

(c) Identify which analysis has the best true ability to resolve two nearby sinusoids.

(d) Explain why zero-padding produces a smoother-looking spectrum without creating new information.

(e) State one practical benefit of zero-padding.

(f) State one incorrect conclusion that could be drawn from the smoother plot.

11. Spectrum of an FIR filter

The impulse response of an FIR filter is $h[n] = \{1,2,3,2,1\}$.

(a) Compute its five-point DFT.

(b) Compute a 64-point DFT after zero-padding the impulse response.

(c) Plot both magnitude responses against normalized radian frequency.

(d) Explain why the 64-point result appears smoother.

(e) State whether the filter itself has changed.

(f) Determine the DC gain directly from $h[n]$ and verify it in the DFT.

(g) Explain how increasing the actual impulse-response length can sharpen filter features, whereas increasing only the DFT length cannot.

12. Periodic signals and coherent sampling

Consider the periodic discrete-time signal $x[n] = 2 + \cos\left(\frac{\pi n}{4}\right) + 0.5\cos\left(\frac{\pi n}{2} + \frac{\pi}{3}\right)$.

(a) Determine the fundamental period.

(b) Compute the DFT of exactly one period.

(c) Identify the bins containing nonzero coefficients.

(d) Relate the DFT coefficients to the Fourier-series coefficients.

(e) Use the inverse DFT to reconstruct one period.

(f) Predict what happens if a record containing 1.5 periods is transformed.

(g) Explain why selecting an integer number of periods avoids artificial boundary discontinuities and leakage.

13. FFT computational advantage

A direct N-point DFT requires approximately N^2 complex multiplications, whereas a radix-2 FFT requires approximately $\frac{N}{2}\log_2 N$.

(a) Calculate both operation counts for $N = 64$, $N = 1024$, and $N = 65\,536$.

(b) Determine the ratio between the direct-DFT and FFT counts in each case.

(c) Explain why the advantage becomes more significant as N increases.

(d) Describe the even-odd decomposition used by a radix-2 FFT.

(e) Explain what is reused in the FFT butterfly structure.

(f) Clarify the statement: “The FFT is not a different transform from the DFT.”

14. Constructing and calibrating a spectrogram

A signal is sampled at $f_s = 16\,000$ samples/s. A spectrogram is calculated using window length $L = 400$, FFT length $N = 1024$, and hop size $R = 160$.

(a) Determine the window duration in milliseconds.

(b) Determine the time separation between consecutive frames.

(c) Calculate the percentage overlap.

(d) Determine the DFT-bin spacing in hertz.

(e) Using the approximation $\Delta f_{\text{Hamming}} \approx \frac{4f_s}{L}$,

(f) estimate the effective frequency resolution.

(g) Explain why DFT-bin spacing and effective frequency resolution are not the same.

(h) Determine the number of positive-frequency bins normally displayed for a real signal.

(i) Describe the effect of replacing $L = 400$ by $L = 1600$ while keeping the hop duration proportional to the window length.

15. Spectrogram interpretation and analysis design

A recording contains five successive events:

- a 500-Hz pure tone lasting 0.5 s;

- a linear chirp rising from 500 Hz to 3000 Hz over 1 s;
- a periodic musical note with fundamental frequency 250 Hz and five harmonics;
- a short broadband impulse;
- a voiced speech segment.

For each event:

(a) Sketch or describe its expected spectrogram.

(b) Identify whether horizontal lines, diagonal ridges, harmonic stacks, broadband vertical structures, or formant regions should appear.

(c) Explain how a long analysis window would affect the display.

(d) Explain how a short analysis window would affect the display.

(e) Choose an appropriate window length and overlap for distinguishing the events.

(f) Explain whether linear magnitude or decibel magnitude would be more informative.

(g) Describe one feature that could be misinterpreted as noise or as an additional component because of spectral leakage.

Finally, write a short spectrum-analysis workflow covering sampling-rate verification, segment duration, window selection, FFT length, frequency-axis calibration, amplitude scaling, and validation with known signals.